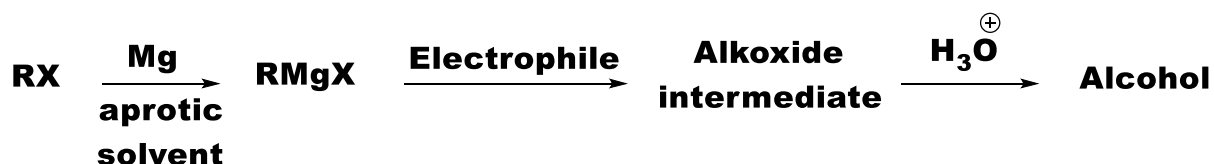
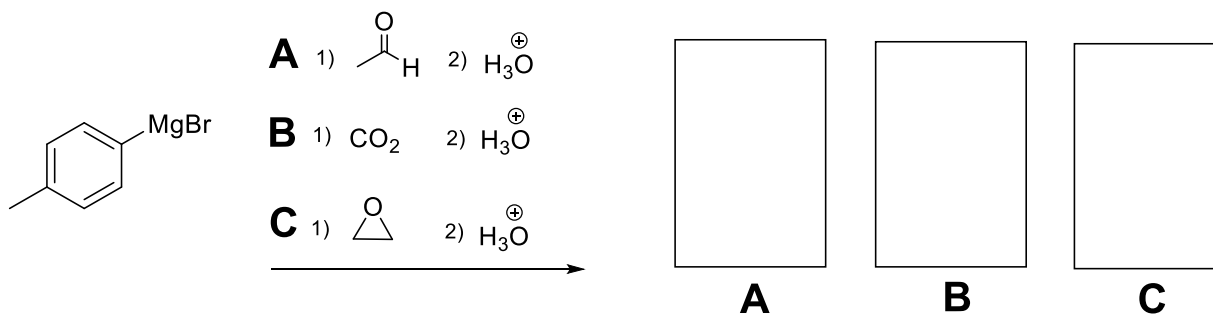


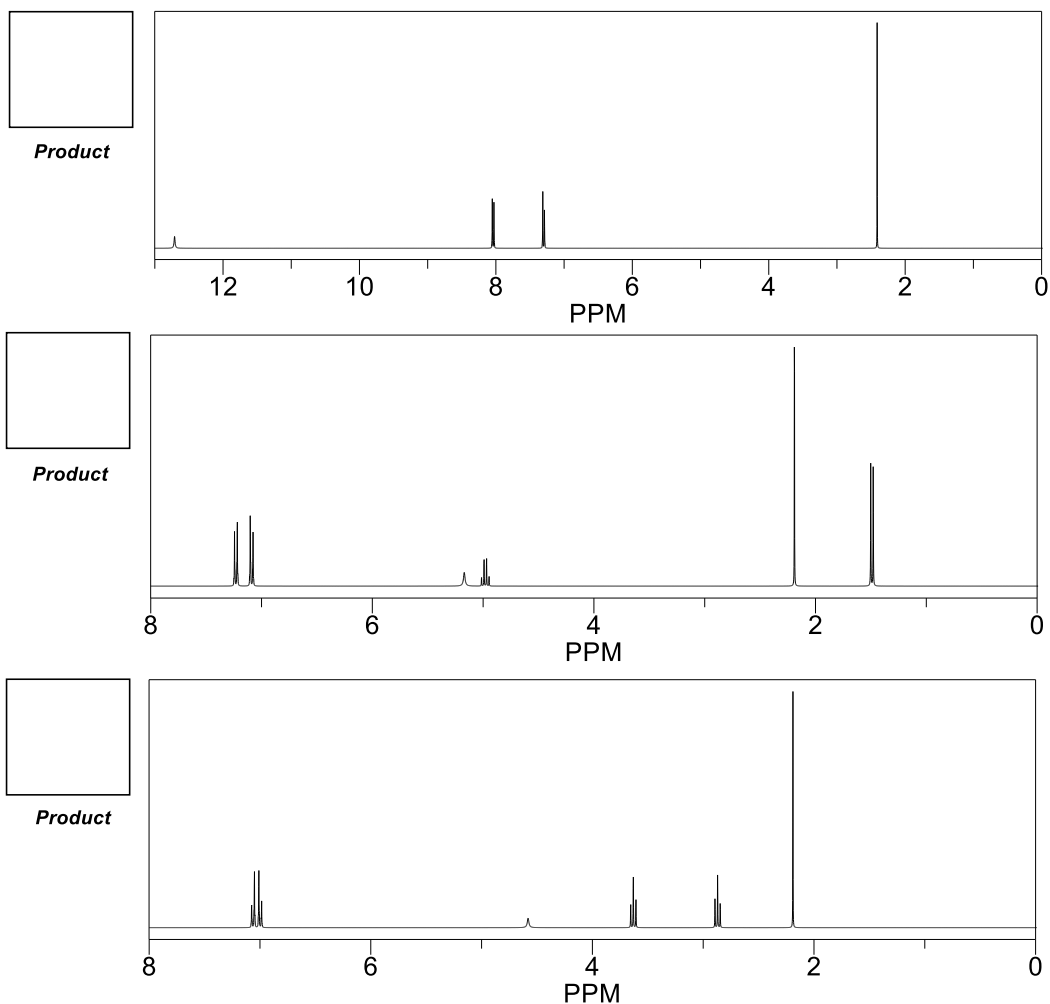
Final Review

Activity 7: Grignard Reaction

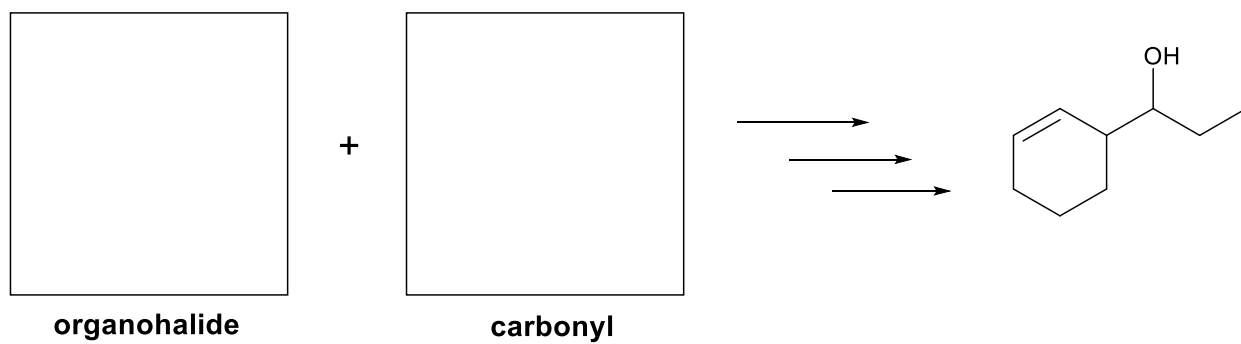


1. What can the halogen be in the organohalide?
2. Why must an aprotic solvent be used?
3. Draw the expected product for each reaction of the following Grignard reagents with conditions A, B, and C. Then match each product with the corresponding NMR spectrum.





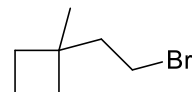
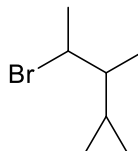
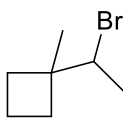
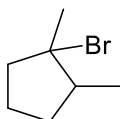
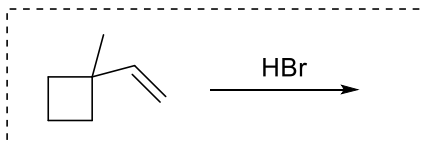
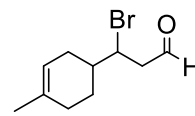
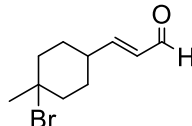
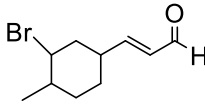
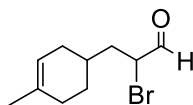
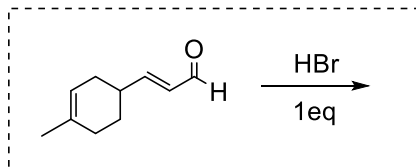
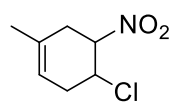
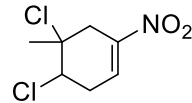
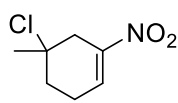
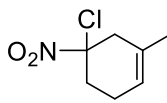
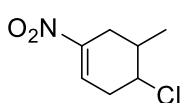
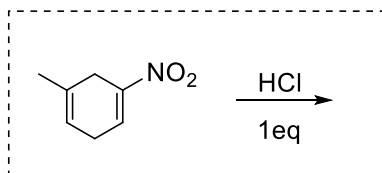
4. Draw possible starting materials that could be used to synthesize the alcohol.



Activity 8: Alkene Addition

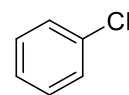
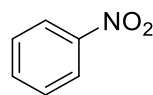
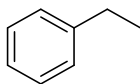
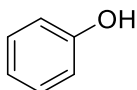
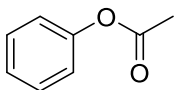
- **Regioselectivity follows the Markovnikov rule.**
- **Chemoselectivity involves the reaction occurring at the more nucleophilic double bond if multiple double bonds are present.**

1. Circle the expected MAJOR product of each of the following alkene addition reactions.

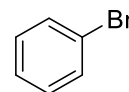
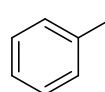
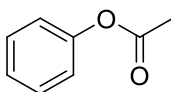
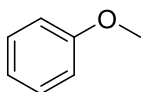
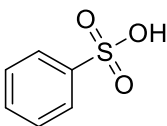


Activity 9: EAS Reactions

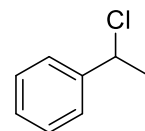
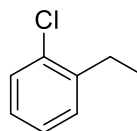
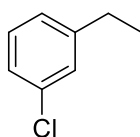
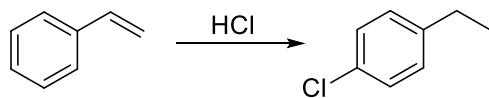
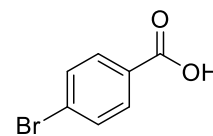
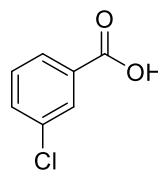
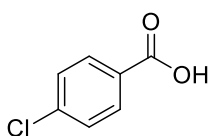
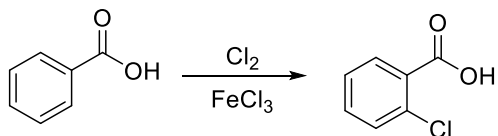
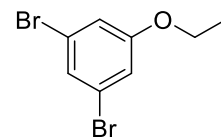
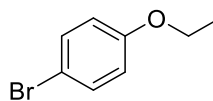
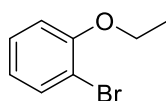
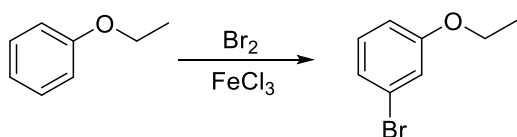
1. Circle the benzene derivative that is the most activated towards electrophilic aromatic substitution.



2. Circle the benzene derivative that is least activated towards electrophilic aromatic substitution.



3. Circle the expected major product of each of the following reactions.



★ The following pages
include old written
quizzes and exams from
the pre-canvas days.
Note that your exam
will NOT have this format.
Use for practice.

Name: _____

Peoplesoft #: _____

Lecture Instructor: William J. Conroy III

Please read all directions and questions carefully and be sure to legibly mark answers in the designated areas. Illegible answers will be marked incorrect. Partial credit will be given where applicable

Part 1 Matching (30)**Part 2 True/False (30)****Part 3 Misc. (32)****Part 4 Reactions (8)****Total: _____(100)**

Part 1: Matching Match the prompts below with the letter of the term that fits *best*. Some terms will not be used. (30 points total)

- | | |
|------------------------------|------------------------------------|
| a. Integration | k. R_f value |
| b. Hyperconjugation | l. Lithium diisopropyl amide (LDA) |
| c. Dehydration | m. Chemically active extraction |
| d. Thin layer chromatography | n. Reverse Phase Chromatography |
| e. Recrystallization | o. Chemical Shift |
| f. Solid-liquid extraction | p. Resonance |
| g. Grignard reagent | q. Electrophile |
| h. Sodium hydroxide | r. Condensation |
| i. Eluent | s. Melting Point |
| j. Distillation | t. Partition coefficient (K_p) |

_____ Expected to be small for polar molecules on silica

_____ Value effected by the chemical environment of a proton

_____ Another name for solvents employed in chromatography

_____ The reason that the conjugate base of phosphoric acid is not a strong nucleophile

_____ Technique involving C-18 modified silica

_____ Associated with quantity of chemically equivalent protons in H-NMR

_____ Decreased in the presence of impurities

_____ Technique used to purify solids

_____ Nucleophilic source of carbon used as a reducing reagent

_____ Process used to separate two compounds with similar polarity, by altering one

_____ Common technique used to monitor the progress of a reaction

_____ Process resulting in alkene formation through the removal of water

_____ Technique used to separate liquids due to differences in boiling point

_____ Common reagent used to achieve non-equilibrium formation of an enolate

_____ Process that results in the loss of water by the joining together of two molecules

Part 2: True or False Determine if the following statements are true or false. (30 total)

_____ Solvent moves by gravity column chromatography

_____ When performing a liquid-liquid extraction with an organic solvent and water, the organic layer is always the top layer

_____ Extraction solvent should be miscible with water

_____ A nitrile (-CN) substituent is a *meta* director in EAS reactions

_____ Phenol is a good solvent for a Grignard reaction

_____ A solvent used for recrystallization should have a boiling point lower than the solids melting point

_____ The rate of E2 reaction is dependent on the rate of carbocation formation

_____ Aldehydes are generally more electrophilic than ketones

_____ 1-Fluorobutane would be a good reactant to form a Grignard reagent

_____ A chloro substituent is a meta director in EAS reactions

_____ Three-bond coupling constants are generally larger than four-bond coupling constants in aromatic compounds

_____ Capillary action moves the mobile phase when developing a TLC plate

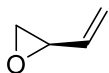
_____ Non-nucleophilic strong acids help facilitate E1 reactions over SN1 reactions

_____ Compounds with a small R_f value have a higher affinity for the stationary phase in normal-phase chromatography

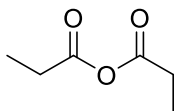
_____ A compound with a small K_p has a higher affinity for silica

Part 3: Miscellaneous

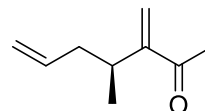
3.1) (6 points) How many unique signals would be observed in an H-NMR spectrum of each of the compounds below? Assume a high enough resolution such that each signal is clearly distinguishable. You can assume that diastereotopicity is observed if applicable.



answer

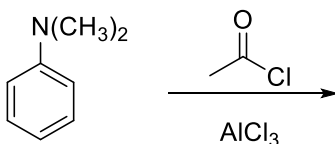


answer



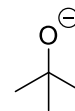
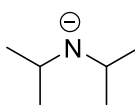
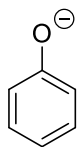
answer

3.2) (2 points) What is the purpose of the aluminum chloride in the following reaction? (*Circle one*)

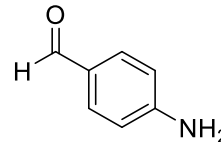
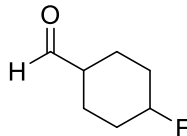
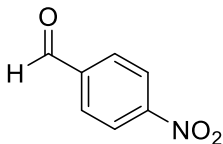
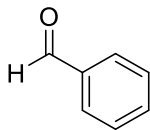


- A. It makes the aromatic substrate more nucleophilic
- B. It transforms the acyl chloride into a more nucleophilic species
- C. It makes the aromatic substrate more reactive towards nucleophiles
- D. It transforms the acyl chloride into a more electrophilic species

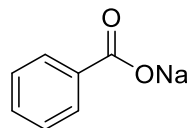
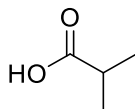
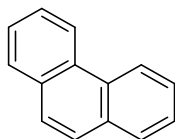
3.3) (2 points) Circle the strongest base from the series below:



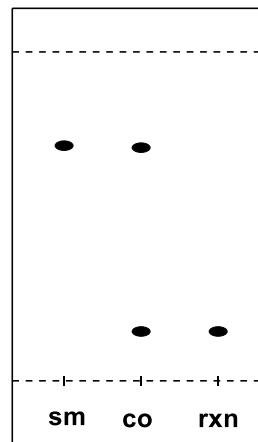
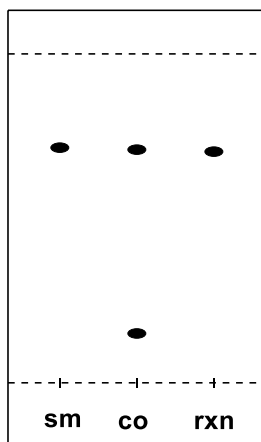
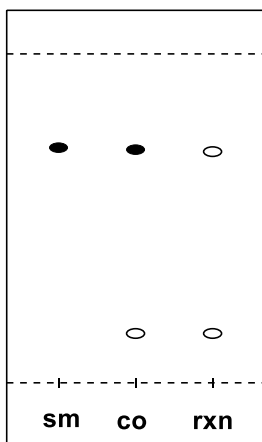
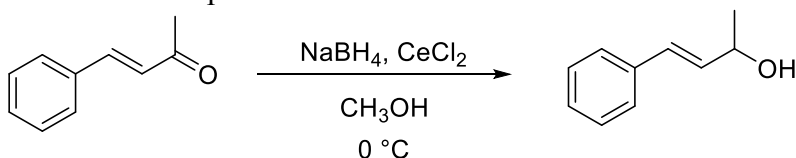
3.4) (2 points) Circle the most electrophilic aldehyde from the series below:



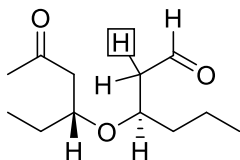
3.5) (4 points) An ethereal solution containing the three compounds below is treated with 3 M aqueous NaOH. Circle the compound(s) below that are found in the aqueous layer after treatment.



3.6) (4 points) For the following reaction, TLC analysis was performed to monitor the progress of the reaction, using a silica plate. Visualization of the organic compound spots on TLC can be made upon exposure of each solvent-dried plate to a UV lamp. On each TLC plate, a student has placed a sample of the starting material (sm) as a reference on the left of the plate, a spot of the reaction mixture (rxn) on the right, and a co-spot (co) in the center of each. Circle the plate that represents the reaction at 50% completion.

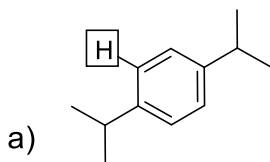


3.7) (2 points) What is the expected splitting pattern of the hydrogen indicated by a box in the molecule below? You can assume only 2- and 3- bond couplings are observed and that none of the coupling constants are equal.

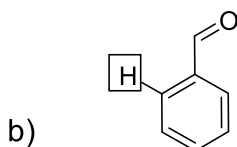


- A. Singlet
- B. Doublet
- C. Doublet of doublets
- D. Doublet of doublet of doublets
- E. Triplet

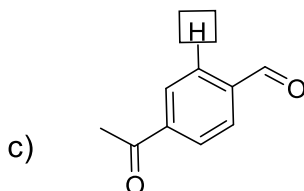
3.8) (6 points) Assuming aromatic 3- and 4- coupling is observed and $J_{3\text{-bond}} > J_{4\text{-bond}}$, what is the expected splitting pattern of the hydrogen indicated by the box in each of the molecules below?



- A. Singlet
- B. Doublet
- C. Triplet
- D. Double of doublets
- E. Triplet of doublets

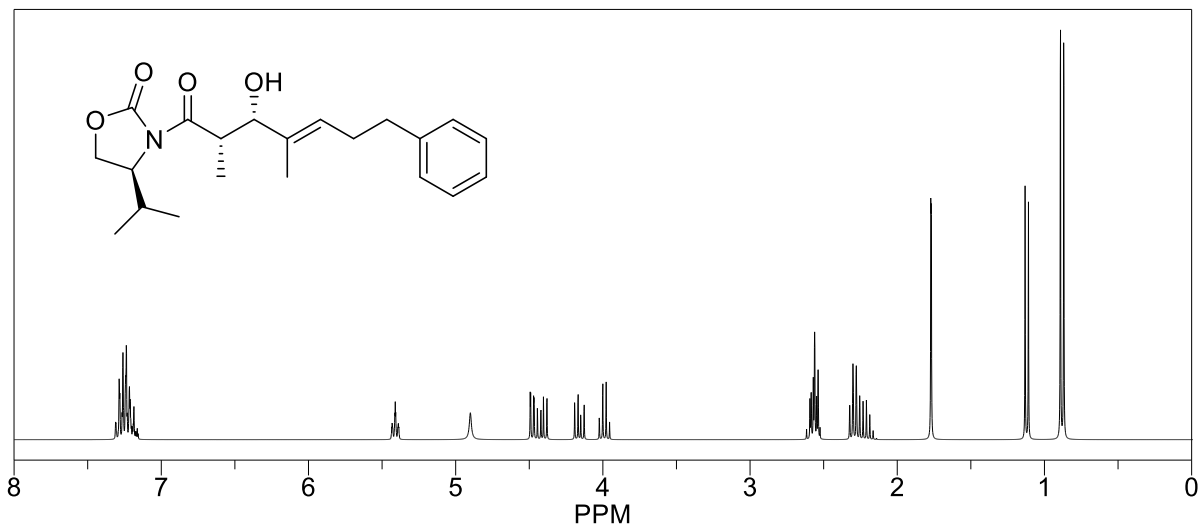


- A. Singlet
- B. Doublet
- C. Triplet
- D. Double of doublets
- E. Triplet of doublets

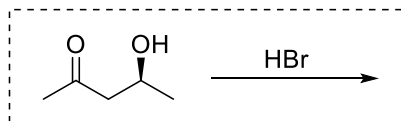
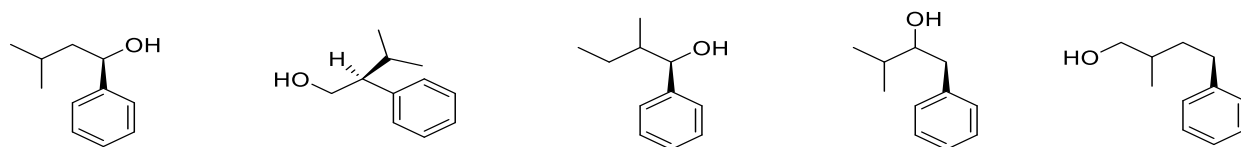
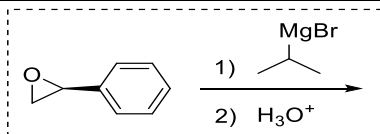
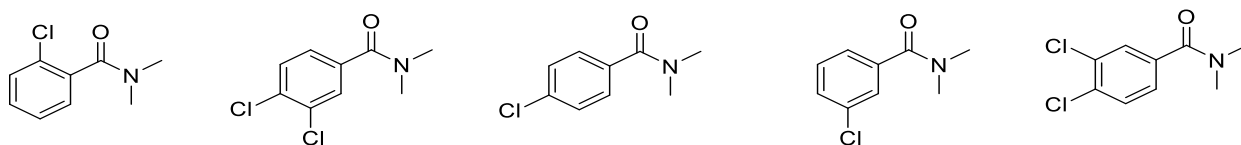
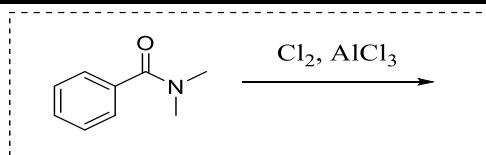
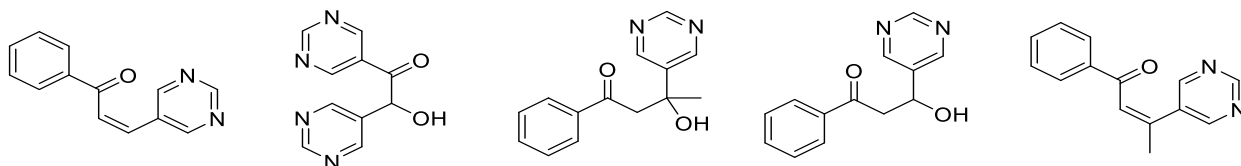
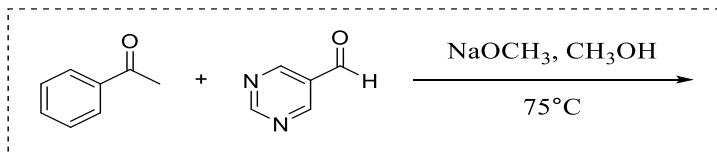


- A. Singlet
- B. Doublet
- C. Triplet
- D. Double of doublets
- E. Triplet of doublets

3.10) (4 points) Circle the peak on the spectrum below that represents the hydrogen atom bonded to the alkene.



Part 4: Reactions (8 points) For each reaction depicted below, circle the expected major product.



Name _____

Lecture Quiz 6

1.) (1 point) When cyclopentanol undergoes an acid-promoted dehydration, it loses a molecule of what?

- A. Acid
- B. Carbon dioxide
- C. Water
- D. Oxygen
- E. Phosphoric acid

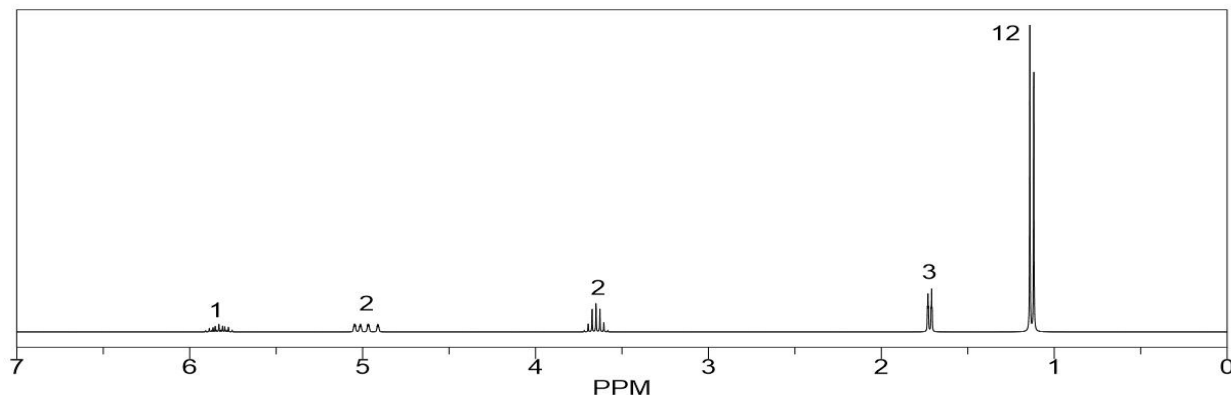
2.) (1 point) A bimolecular reaction where proton abstraction and leaving-group departure are simultaneous is an example of a(n):

- A. E1 reaction
- B. SN2 reaction
- C. E2 reaction
- D. Dehydration
- E. Condensation

3.) (1 point) All of the following are features of an E1 reaction except for:

- A. The rate of an E1 reaction is dependent on the concentration of the base
- B. The mechanism of an E1 reaction involves a carbocation intermediate
- C. An E1 reaction requires a good leaving group
- D. Does not utilize strong bases like LDA

4.) (2 points) A student attempted a dehydration of isopropanol using a low concentration of sulfuric acid as a catalyst. Afterwards an HNMR experiment of the product was performed. The HNMR spectrum provides evidence of a successful dehydration but **also** shows the presence of an undesired side product. The HNMR is shown below.



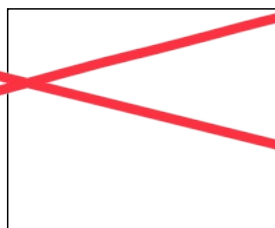
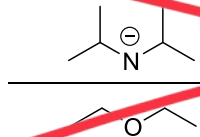
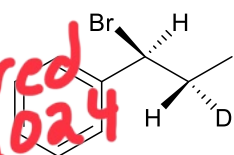
a) Draw the structure of the undesired side product in box provided.



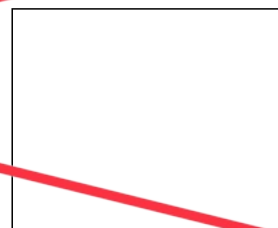
b) What could be changed about the experiment to prevent the formation of this side product?

- Use a strong base instead of acid
- Use a nucleophilic acid instead of sulfuric acid
- Use a low concentration of phosphoric acid
- Use a high concentration of sulfuric acid or phosphoric acid

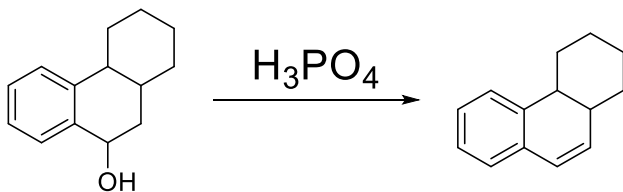
5.) (2 points) Draw the two possible elimination products of the following E2 reaction. Correct stereochemistry (E/Z) must be shown for full credit.



+



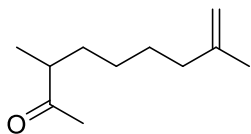
6.) (3 points) Draw the complete electronic mechanism (ARROWS) of the following reaction. Be sure to show ALL bonds being FORMED or BROKEN. Be sure to show formal charge on the appropriate atom(s). You may use H_3O^+ or H^+ as the acid in the mechanism.



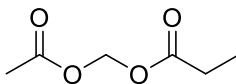
Name _____

Lecture Quiz 7

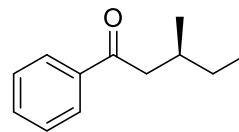
1.) (3 points) How many unique proton chemical environments are in each of the following molecules?



_____ *answer*

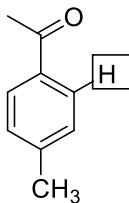


_____ *answer*

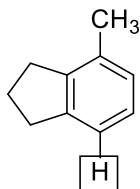


_____ *answer*

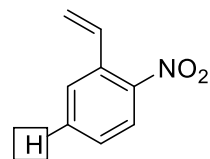
2.) (3 points) What is the expected splitting pattern of the indicated proton in each of the following molecules? Assume $J_{2\text{-bond}} > J_{3\text{-bond}} > J_{4\text{-bond}}$



_____ *answer*

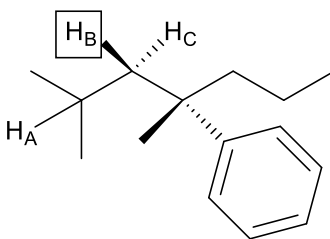


_____ *answer*



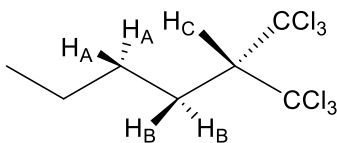
_____ *answer*

3.) (1 point) What is the expected splitting pattern of the indicated proton in the following molecule if $J_{BC} = 14$ Hz and $J_{BA} = 6$ Hz?



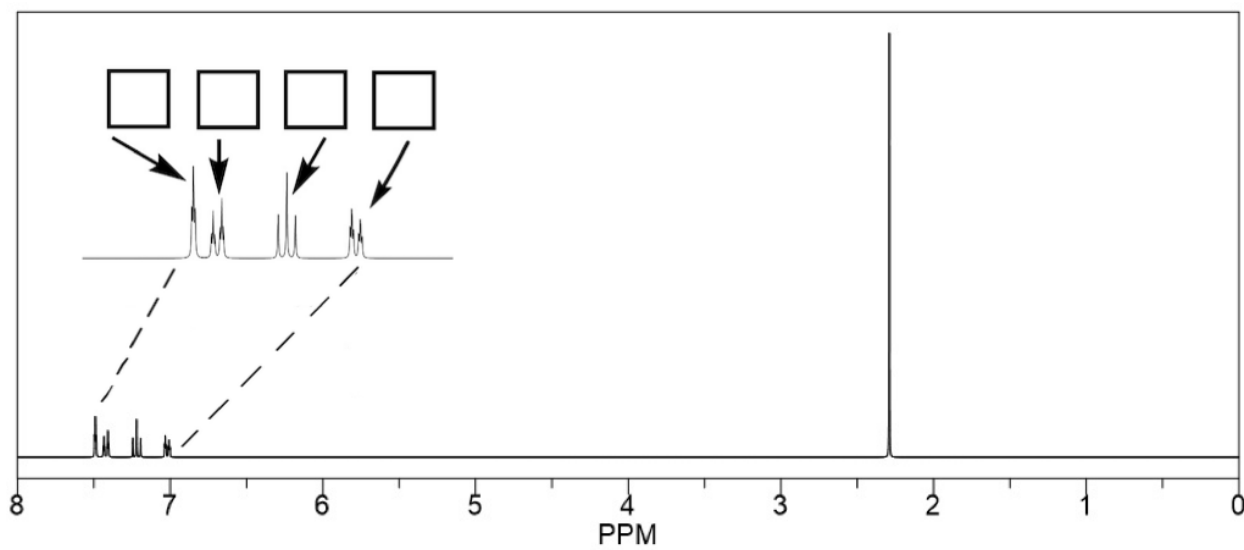
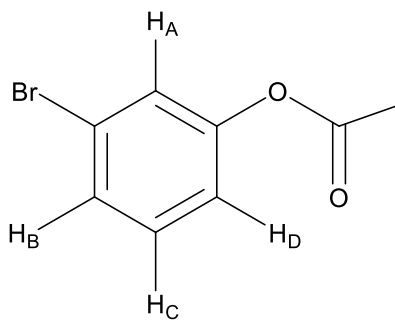
_____ *answer*

4.) (1 point) What is the expected splitting pattern of H_B in the following molecule if $J_{BA} = 7$ Hz and $J_{BC} = 7$ Hz?



answer

5.) (2 points) Assign the aromatic protons (labeled A, B, C, D) of the molecule below to the following spectrum by writing the letter in the box above the corresponding peak.



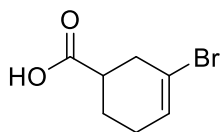
Name _____

Lecture Quiz 8

1.) (1 points) Which of the following would be an appropriate solvent for performing a carbonyl reduction of an aldehyde using a Grignard reagent?

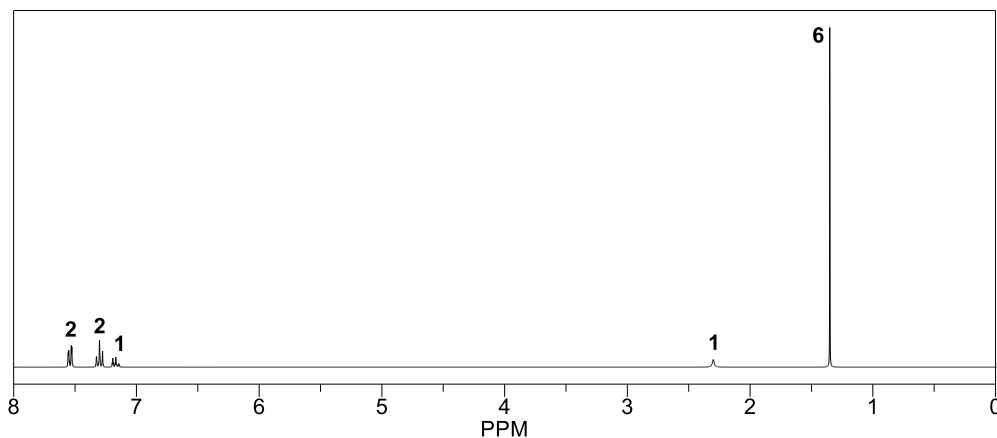
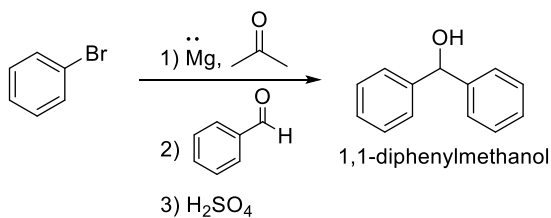
- A. Ammonia
- B. Isopropyl alcohol
- C. Tetrahydrofuran
- D. Water
- E. Ethyl Acetate

2.) (1 points) A student attempted to prepare a Grignard reagent from the organohalide shown below. The reaction was not successful because:



- A. The carbon-bromine bond is too strong
- B. Grignards cannot form on sp^2 hybridized carbon atoms
- C. The alcohol will get oxidized into a ketone by Mg
- D. The hydroxyl proton is too acidic

3.) A student attempted to synthesize 1,1-diphenylmethanol using the following series of reactions involving the formation of phenyl magnesium bromide in acetone followed by the addition of benzaldehyde. Upon completion the student performed an HNMR experiment. The spectrum did not support that the reaction was a success. The spectrum showed evidence of a different product.



- a) (1 point) Draw the structure, in the box below, of the undesired product represented by the NMR spectrum.



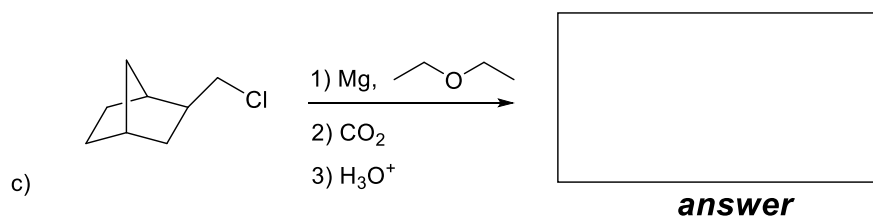
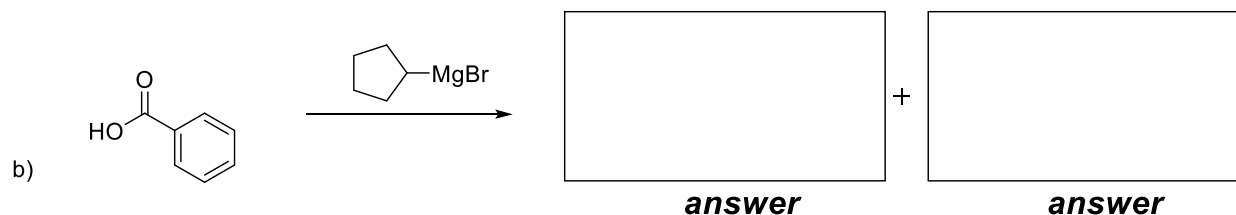
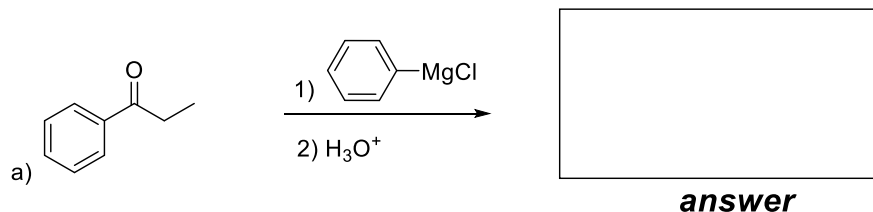
answer

- b) (1 point) What could be changed about the experimental procedure in order to achieve a successful synthesis of 1,1-diphenyl methanol? What would you use to correct it? (*one very brief statement in approximately 5 words or less*)

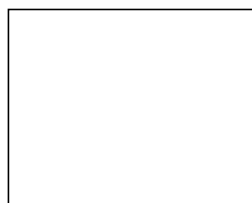


answer

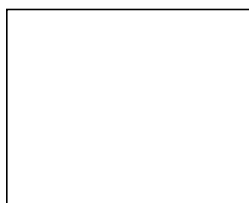
- 4.) (4 point) Determine the product(s) of each of the following reactions by drawing the chemical structure in the boxes. No stereochemistry needs to be shown.



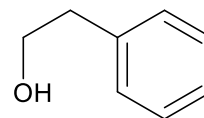
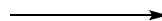
5.) (2 points) Determine the organohalide (R-X) and electrophile (aldehyde or ketone) that can be used in a Grignard reaction to form the given product. Assume there is an acidic workup step. *(Draw the organohalide, not the Grignard. You cannot use "R" for short in either structure).*



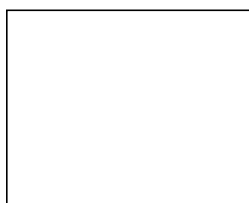
organohalide



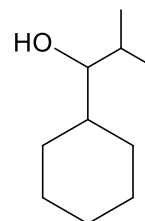
electrophile



organohalide



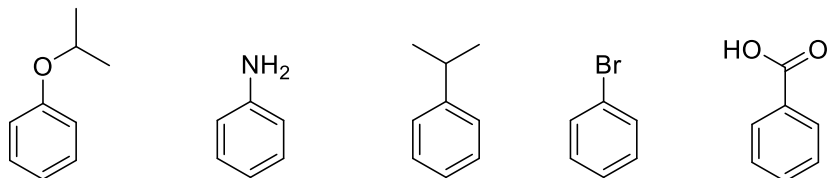
electrophile



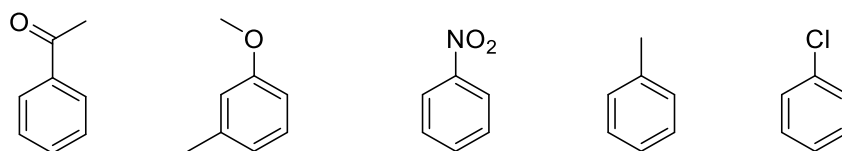
Name _____

Lecture Quiz 9

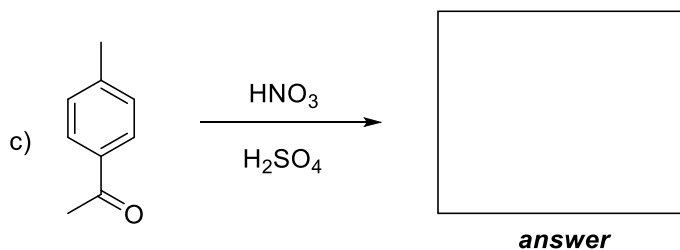
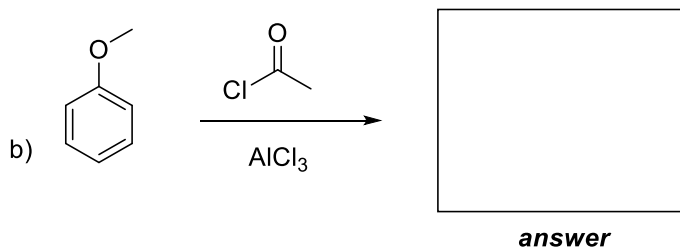
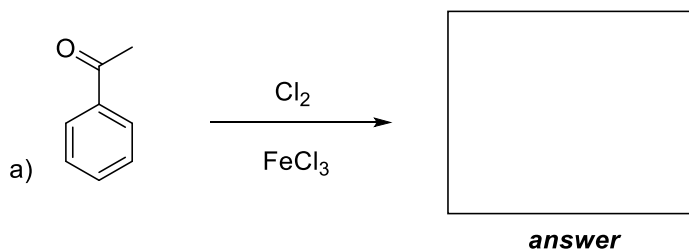
1.) (1 points) Circle the benzene structure that would undergo an electrophilic aromatic substitution reaction the slowest.

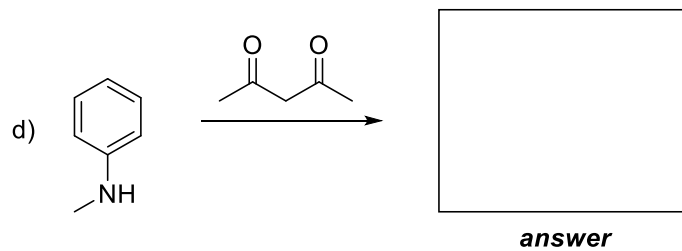


2.) (1 points) Circle the benzene structure that is most activated towards electrophilic aromatic substitution.

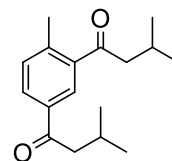
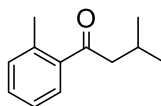
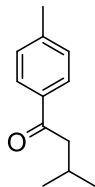
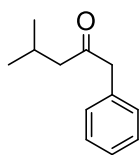
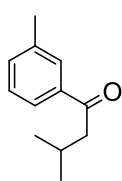
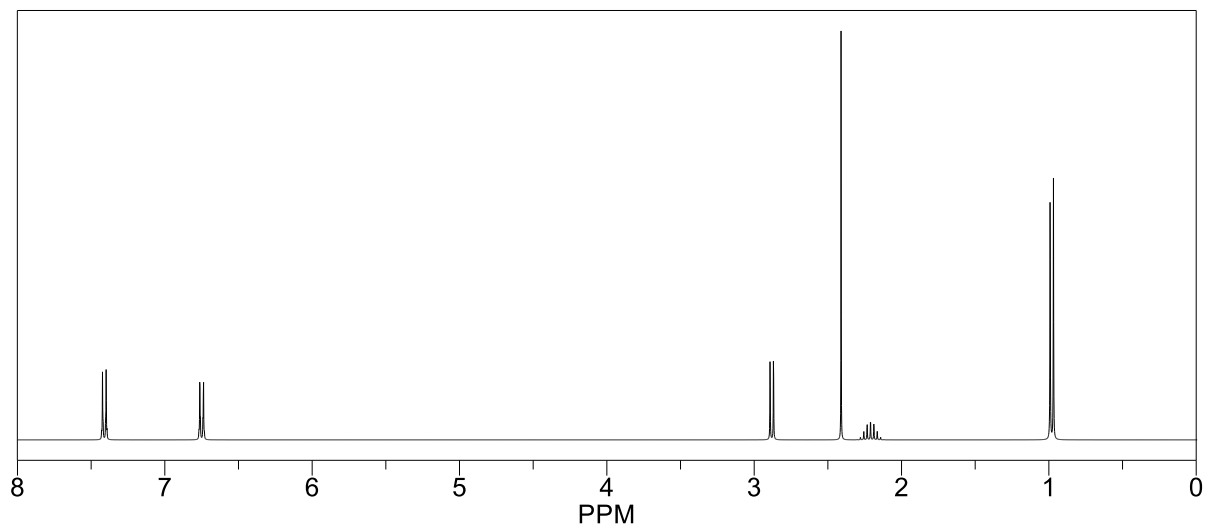
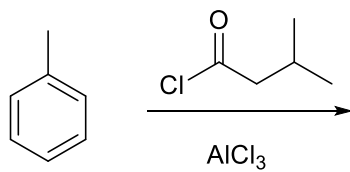


3.) (3 points) Draw the MAJOR product of each of the following reactions. (*Please be sure to draw your final answer in the box*)

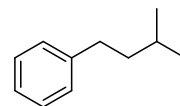
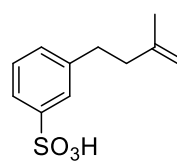
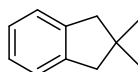
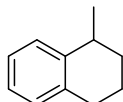
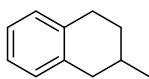
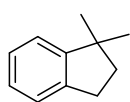
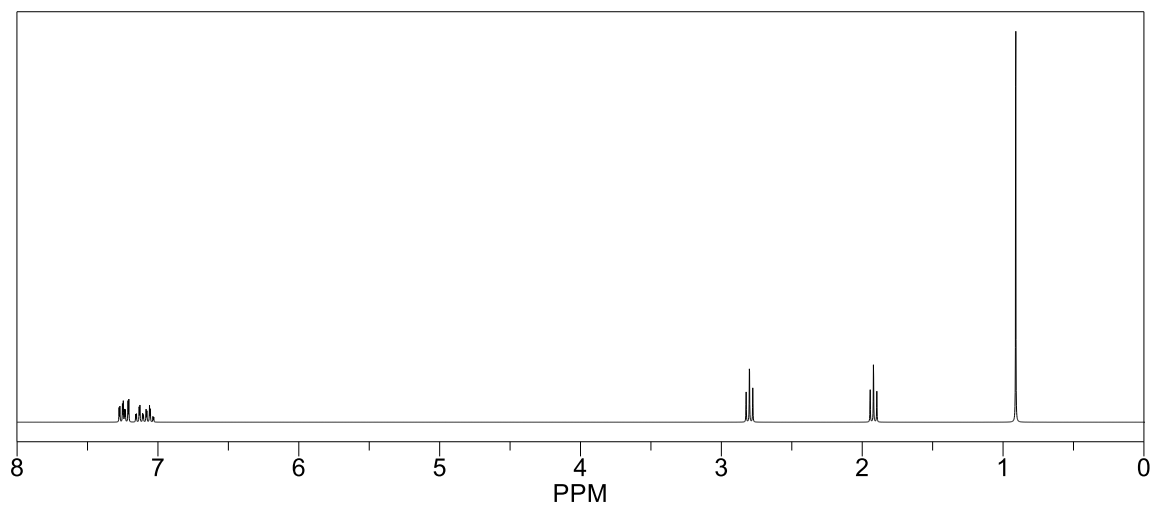
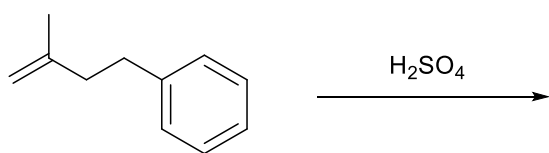




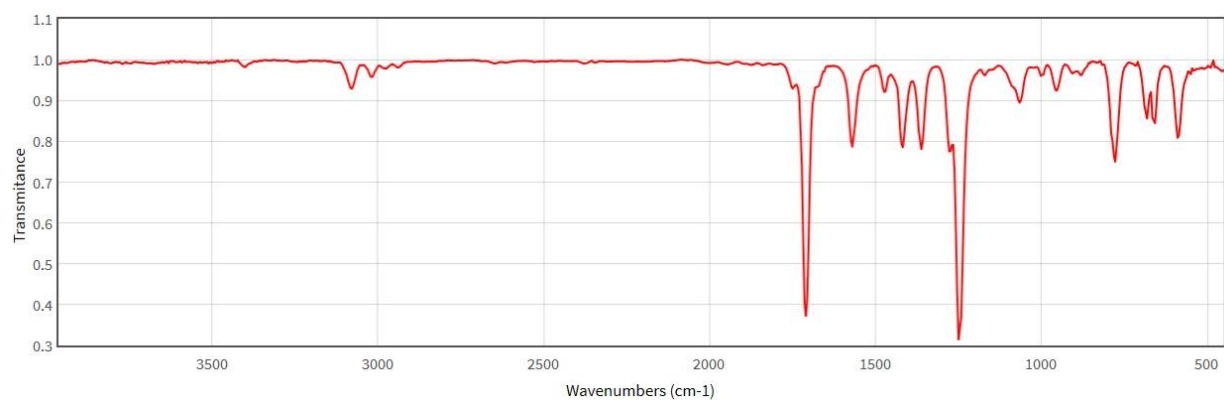
4.) (1 point) Based on the H-NMR below, circle the product of the reaction shown below:



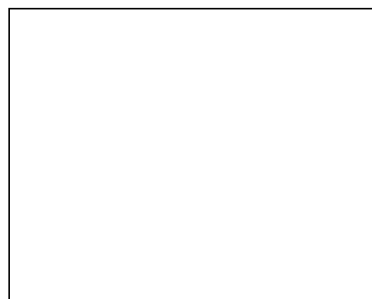
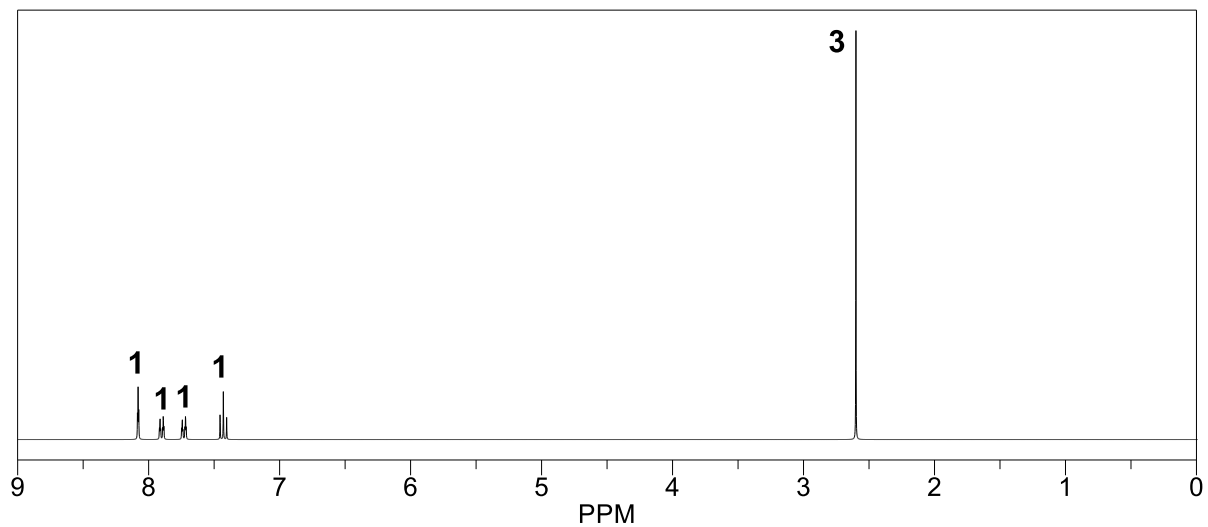
5.) (1 Point) Based on the H-NMR below, circle the product of the reaction shown below:



6.) (2 Points) Determine the molecular structure of a molecule with the formula C_8H_7BrO based on the following IR spectrum and H-NMR spectrum. (*please draw your final structure in the box provided*)



C_8H_7BrO



answer